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PHOTometry and spectrometry EXTRAction code: multiwavelength photo-spec pipeline to merger and no merger galaxies

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ABSTRACT

Context. Spectral energy distribution (SED) studies of galaxies in nearby clusters increasingly require the consistent combination of multi-survey imaging photometry with single-fibre optical spectroscopy. For interacting and merging pairs, ground-based seeing and the broad mid-infrared point spread functions (PSFs) blend the light of the two components, biasing both the photometry and any derived physical parameters.

Aims. We present `photextra_pipeline`, an open-source Python pipeline that automates, for arbitrary lists of targets, the retrieval of ultraviolet to mid-infrared imaging (nine independent surveys, 21 bands) and optical spectra, component-level deblending of close galaxy pairs, homogenised aperture photometry, stellar+AGN spectral fitting, and a physically motivated combination of the photometric and spectroscopic products in a single reproducible flow.

Methods. From only (RA, Dec, z, ID) per target, the pipeline downloads GALEX, DESI Legacy Surveys, and WISE cutouts, separates the components of close pairs with the companion package `xdebpair`, measures per-component isolated-image photometry with PSF homogenisation to the WISE W4 resolution, resolves and fits the matching DESI or SDSS spectrum with the companion package `XpectraFit` (pPXF + E-MILES stellar continuum plus an optional AGN component), and rescales the fibre spectroscopy to total light with a wavelength-dependent aperture correction anchored on the imaging photometry, extending the grey-factor approach of Zou et al. (2024).

Results. We validate the pipeline on a controlled set of five close systems spanning the single/companions/merger classification space, and on a production run of 450 spectroscopic members of the galaxy cluster MKW 8 ($z \simeq 0.027$) from the CHANCES survey, of which 446 (99.1 %) yield complete combined photometry+spectroscopy products, at a median cost of ≃220 s per target end to end and <1 s when resuming from cached products.

Conclusions. `photextra_pipeline` provides a fully configuration-driven, checkpointable route from target coordinates to science-ready, component-level SEDs and aperture-corrected spectroscopic parameters, and is directly applicable to wide-field cluster surveys such as CHANCES.

Key words. methods: data analysis – techniques: photometric – techniques: spectroscopic – galaxies: photometry – galaxies: interactions – galaxies: clusters: general

1. Introduction

The physical characterisation of galaxies through their spectral energy distributions (SEDs) rests on the availability of homogeneous multi-band photometry, and, whenever possible, on its combination with optical spectroscopy that constrains the stellar populations, emission-line physics and nuclear activity of the same objects (e.g. Boquien et al. 2019; Zou et al. 2024). Modern public surveys make the raw ingredients abundant: the DESI Legacy Imaging Surveys (Dey et al. 2019) provide deep optical grz imaging over ~20 000 deg^2, GALEX (Martin et al. 2005) and WISE (Wright et al. 2010) bracket the optical with ultraviolet and mid-infrared coverage, and the DESI and SDSS spectroscopic programmes (DESI Collaboration et al. 2024; York et al. 2000) supply fibre spectra for millions of galaxies. Yet assembling these ingredients into science-ready, per-galaxy SEDs remains laborious, and three practical problems recur.

First, *resolution heterogeneity*: the PSF full width at half maximum (FWHM) ranges from ≃1.3″ in the Legacy Surveys optical bands to 12″ in WISE W4, so consistent total-flux measurements require PSF homogenisation, careful aperture definitions, and per-survey zero-point handling. Second, *blending*: for interacting and merging pairs – a population of particular interest in dense environments – the projected nuclear separations (≲10″) are comparable to or smaller than the PSF of most of the bands involved, so the components must be deblended at the pixel level before any per-component photometry is meaningful. Third, *aperture mismatch*: fibre spectrographs capture only the central 1.5″ (DESI) or 3″ (SDSS) of a nearby galaxy, so spectroscopically derived quantities such as stellar masses refer to a small and wavelength-dependent fraction of the total light and must be tied to the imaging photometry before they can be compared with photometric SED fits (Zou et al. 2024).

General-purpose tools exist for individual steps – source extraction (Bertin & Arnouts 1996; Barbary 2016), PSF photometry, SED fitting (Boquien et al. 2019), full-spectrum fitting (Cappellari 2017) – but, to our knowledge, no public pipeline chains *component-level deblending of close pairs*, *multi-survey aperture photometry*, *stellar+AGN spectral fitting* and a *consistent photometry–spectroscopy combination* into a single reproducible, configuration-driven flow. `photextra_pipeline` was developed to close this gap in the context of the CHANCES cluster survey (Sifón et al. 2025), where thousands of spectroscopic cluster members – among them close pairs and ongoing mergers – must be processed uniformly.

This paper describes the design and implementation of `photextra_pipeline` and its two scientific companion packages, `xdebpair` (pair deblending) and `XpectraFit` (spectral fitting), and validates the full system on real data. Section 2 summarises the input data. Section 3 presents the architecture and the three operation modes. Sections 4–6 describe the photometric, spectroscopic and combination stages. Section 7 reports the validation on a controlled five-system test set and on a 450-galaxy production run on the cluster MKW 8. Section 8 details code availability.

2. Input data

2.1. Imaging

The pipeline supports the 21 ultraviolet-to-mid-infrared bands listed in Table 1, spanning nine independent surveys, all of them optional and selectable per run through the target configuration. Optical g, r, z cutouts are retrieved from the DESI Legacy Imaging Surveys DR10 cutout service (Dey et al. 2019) at the native $0.262'' \, \mathrm{px}^{-1}$ scale, with an exponential-backoff retry policy against rate limiting; a bulk-download alternative for this survey is described below. GALEX FUV/NUV and WISE W1–W4 cutouts are retrieved through `SkyView` (McGlynn et al. 1998) via `astroquery` (Ginsburg et al. 2019), requested explicitly at each survey's native pixel scale so that the tabulated zero points apply (Sect. 4.3). SDSS *ugriz* (York et al. 2000) are likewise retrieved through `SkyView`. Pan-STARRS1 *grizy* (Chambers et al. 2016) are retrieved from the public stack-image cutout service (`ps1images.stsci.edu`), dividing by the exposure time so a fixed AB zero point of 25.0 applies uniformly. 2MASS JHK_s (Skrutskie et al. 2006) are retrieved from the IRSA image-cutout service, rescaling every per-image `MAGZP` header value to a common Vega zero point before applying the Vega→AB offset, because we found the `SkyView` 2MASS mirror keeps the native per-image zero point, which a single fixed value in the registry cannot represent (a field-to-field scatter of 0.5 mag was measured across three test fields). Spitzer IRAC1–4 are retrieved from the public Spitzer Enhanced Imaging Products (SEIP) mosaics via the IRSA Simple Image Access service; because SEIP mosaics carry physical surface-brightness units (MJy sr^{-1}) rather than raw counts, their zero point follows directly from the pixel solid angle rather than from any catalog offset (Table 1, note *c*). SEIP coverage is restricted to targeted Spitzer fields ($\lesssim$ a few percent of the sky), so most random sky positions legitimately return "no coverage" for this survey alone – a per-target coverage table and footprint map are produced by the companion `survey_coverage` module (Sect. 2.2) before committing to a large run.

All survey-specific properties (effective wavelength, PSF FWHM, pixel scale, photometric zero point) are centralised in a single registry (`SURVEY_DEFAULTS` in `downloader.py`); no other module hard-codes survey physics, which is what allowed the five surveys above (SDSS, PS1, 2MASS, Spitzer, plus footprint-only support for further archives, Sect. 2.2) to be added as a registry entry and a retrieval function each, without touching the photometry, deblending or spectroscopy code. The default cutout size is $1'$ (configurable), and every downloaded image is cached on disk as FITS so that re-runs are network-free.

2.2. Coverage checking and non-blind bulk imaging

Two further archives are supported for *coverage discovery only*, without a general-purpose per-target cutout path: the Hubble Space Telescope archive, queried through `astroquery.mast` to report, per target, whether any HST imaging exists nearby and in which instrument/filter (pointed observations only, essentially never covering a random cluster member, but occasionally relevant for special targets); and J-PLUS, whose public SIAP footprint service answers a covered/ not-covered query without authentication even though its bulk images are not exposed through a cutout-friendly interface. Two additional archives with genuine imaging depth over relevant sky area – Hyper Suprime-Cam and S-PLUS – were evaluated and found to require an authenticated account for both cutouts and footprint queries; they are recorded as `unknown (auth required)` rather than guessed. A companion module, `survey_coverage`, runs lightweight metadata queries (never a full image download) for every supported archive over an arbitrary target list and produces a per-target coverage table and a sky-position footprint map, so that a large run can be planned – and archives with partial or non-existent coverage identified – before any imaging is retrieved.

For very large target lists ($\gtrsim 10^5$), the per-target cutout services above become rate-limit-bound rather than bandwidth-bound: at the empirically measured $\simeq$36 s per target for the seven-band default set, downloading cutouts for 3×10^5 targets one at a time would take on the order of weeks. The pipeline therefore also supports an opt-in bulk-download path for the Legacy Surveys imaging, the dominant optical component: instead of one HTTP cutout request per target, the underlying DR10 brick mosaics (fixed 0.25×0.25 deg tiles tesselating the whole footprint) that actually overlap the target list are downloaded once, in bulk, from the plain file server, and each target's cutout is then extracted locally by WCS-based array slicing, no further network calls involved. Because two of our target catalogues (the CHANCES DESI cross-match and the CAVITY/Pan-STARRS-Legacy cross-match) already carry a resolved Legacy brick identifier per target, the set of bricks to fetch is known without any additional spatial indexing. This conversion from $O(N_{\mathrm{targets}})$ rate-limited requests to $O(N_{\mathrm{bricks}})$ bulk transfers is most valuable when targets are spatially clustered (e.g. cluster members around a handful of pointings, where several targets per brick share a single download) and provides little or no benefit when targets are sparse across a large contiguous footprint (close to one target per brick, so nearly the whole brick mosaic would be fetched for a single cutout); the pipeline therefore leaves this path opt-in rather than default, and a target list's brick-sharing ratio is the deciding factor for whether it is worth enabling.

2.3. Spectroscopy

Targets never need to supply a spectrum file. Given (RA, Dec, z), the `spectrum_acquisition` module resolves the matching spectrum automatically: DESI spectra are queried first through the SPARCL client (Juneau et al. 2024), with a fallback to a NOIRLab Astro Data Lab SQL query plus direct HTTPS retrieval from the DESI DR1 healpix coadd tree when SPARCL is unreachable; targets without a DESI match fall back to a coordinate-based SDSS cross-match within a $3''$ radius. Resolved spectra are cached per target (DESI as compact `.npz` arrays of wavelength, flux and inverse variance), so batch re-runs never re-download. All acquisition failures are soft: a missing or unreachable spectrum can never abort a batch run.

Table 1. Imaging bands supported by `photextra_pipeline` and their adopted properties (the `SURVEY_DEFAULTS` registry).

Band	$\lambda_{\rm eff}$ (μm)	PSF FWHM ($''$)	Pixel scale ($''$/px)	ZP[a] (AB mag)
GALEX FUV	0.153	4.5	1.5	18.82
GALEX NUV	0.231	6.0	1.5	20.08
SDSS u	0.355	1.3	0.396	22.5[b]
SDSS g	0.469	1.3	0.396	22.5[b]
SDSS r	0.617	1.3	0.396	22.5[b]
SDSS i	0.748	1.3	0.396	22.5[b]
SDSS z	0.893	1.3	0.396	22.5[b]
Legacy g	0.472	1.3	0.262	22.5
Legacy r	0.641	1.3	0.262	22.5
Legacy z	0.906	1.3	0.262	22.5
PS1 g	0.487	1.3	0.25	25.0[c]
PS1 r	0.622	1.2	0.25	25.0[c]
PS1 i	0.755	1.1	0.25	25.0[c]
PS1 z	0.868	1.1	0.25	25.0[c]
PS1 y	0.963	1.1	0.25	25.0[c]
2MASS J	1.235	2.9	1.0	20.91[d]
2MASS H	1.662	2.9	1.0	21.39[d]
2MASS K_s	2.159	2.9	1.0	21.85[d]
WISE W1	3.4	6.1	1.375	23.199[e]
WISE W2	4.6	6.4	1.375	22.839[e]
WISE W3	12.0	6.5	1.375	23.174[e]
WISE W4	22.0	12.0	1.375	19.620[e]
Spitzer IRAC1	3.550	1.66	0.6	21.582[f]
Spitzer IRAC2	4.493	1.72	0.6	21.582[f]
Spitzer IRAC3	5.731	1.88	0.6	21.582[f]
Spitzer IRAC4	7.872	1.98	0.6	21.582[f]

Notes. [a] AB zero point applying to counts on the native pixel grid, unless noted otherwise. [b] SDSS is natively AB-calibrated; small u, z offsets ($\lesssim 0.04$ mag) are neglected, below the calibration scatter. [c] PS1 stacks are divided by `EXPTIME` so a fixed AB zero point of 25.0 applies (Tonry et al. 2012; Waters et al. 2020). [d] 2MASS Atlas images are Vega-calibrated with a per-image zero point (`MAGZP`); every cutout is rescaled to a common Vega zp of 20.0 before adding the Vega→AB offset (Blanton et al. 2005: $+0.91, +1.39, +1.85$ for J, H, K_s). [e] The WISE entries are the Atlas Vega magnitude zero points converted to AB with the offsets of Jarrett et al. (2011) ($\Delta m = 2.699, 3.339, 5.174, 6.620$ for W1–W4); see Sect. 7.3. [f] Spitzer SEIP mosaics carry physical units (MJy sr^{-1}); the zero point follows from the 0.6$''$ pixel solid angle rather than from a catalog offset.

2.4. Target input

The unit of input is a target dictionary (or CSV row) with `id`, `ra`, `dec`, optional redshift `z` (required for spectroscopy) and an optional morphological `type` column (`merger`, `pre_merger`, `post_merger`, `no-merger`) that steers the separation policy (Sect. 4.1.1). All remaining behaviour is set in a single commented YAML configuration file; Table 2 summarises the main options.

3. Architecture and operation modes

`photextra_pipeline` is organised around a single orchestrator class (`Pipeline` in `pipeline.py`) that consumes the YAML configuration and processes one target at a time; batch drivers simply loop (optionally in parallel) over targets. A `mode` selector chooses among three flows (Fig. 1):

- `photometry`: the full imaging pipeline only – download, source separation, PSF homogenisation, per-component photometry, SED products (Sect. 4);

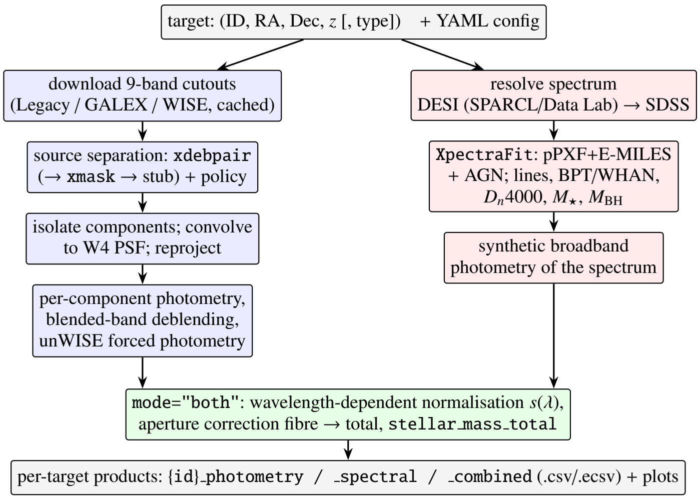

Fig. 1. Architecture of `photextra_pipeline`. Blue boxes belong to the `photometry` mode, red boxes to the `spectroscopy` mode; `both` runs the two branches and the green combination step. Every stage is checkpointed: cached products are reused and only missing pieces are recomputed.

- `spectroscopy`: only spectrum acquisition and the `XpectraFit` spectral fit (Sect. 5); no imaging is touched;
- `both`: the photometric pipeline plus the spectral fit, then the combination step that rescales the fibre spectroscopy to total light using the imaging photometry as absolute reference (Sect. 6), producing one merged `{id}_combined` product per target.

Each target's outputs are written under `{output_dir}/{id}/` in three sub-trees: `cache/` (FITS cutouts), `products/` (machine-readable `.csv`/`.ecsv` tables) and `plots/` (diagnostic figures: per-band processing grids, per-component SEDs, and the combined photometry+spectroscopy plot).

Two engineering choices proved essential for survey-scale runs. First, *checkpointing*: `Pipeline.run()` skips any target whose final product for the selected mode already exists, and `mode="both"` transparently reuses cached `_photometry` and `_spectral` products from earlier passes, computing only the missing pieces and the combination step – resuming an interrupted run, or upgrading a photometry run to a combined one, costs seconds per target (Sect. 7.2). Second, *explicit failure accounting*: any download or query failure is appended to a machine-readable `failed_targets.jsonl` manifest with a stage tag and a reason. Critically, if the Legacy optical imaging that drives the deblending cannot be downloaded, the pipeline refuses to degrade silently to its built-in single-component stub mask and instead marks the target as failed for later re-runs, so a transient rate limit can never masquerade as a science result.

4. Photometric pipeline

4.1. Source separation with `xdebpair`

The default aperture definition (`aperture_mode: mask`) assigns to each galaxy a pixel mask per component. Masks are produced by the companion package `xdebpair`, an iterative deblending code for merging galaxy pairs that operates on the Legacy r-band image (optionally with the g/z bands as a colour cube). `xdebpair` first segments the field (SEP-based segmentation, falling back to marker-controlled

Table 2. Main configuration options of `photextra_pipeline` (YAML schema of `config_xmask.yaml`).

Key	Default	Description
`mode`	`photometry`	Analysis to run: `photometry`, `spectroscopy`, or `both` (Sect. 3).
`surveys`	9 bands	Imaging bands to download and measure (Table 1).
`download_size`	1	Cutout width in arcmin.
`common_grid.pixscale` / `.size`	1.375 / 45	Common WCS grid (arcsec px^{-1} / pixels) used for the visual processing grid; science fluxes are measured on native pixels.
`use_xdebpair`	`true`	Use `xdebpair` for source separation, falling back to `xmask` and then to a circular stub if unavailable.
`photometry.aperture_mode`	`mask`	Aperture definition: per-component pixel masks (`mask`), one fixed circular aperture (`aperture`), or one SEP elliptical aperture (`sep_apertures`).
`photometry.separation`	`pair`	Policy when two components are found: keep the pair (`pair`, classification-aware), measure only the central galaxy (`central`), or one integrated measurement (`total`).
`photometry.aperture_radius_arcsec`	5.0	Radius for `aperture_mode: aperture`.
`sep.thresh_sigma` / `.ellipse_k`	1.5 / 2.5	SEP detection threshold and Kron-like ellipse scale (`sep_apertures` mode).
`sep.ref_survey` / `.max_match_arcsec`	Legacy_r / 10	Detection band and target match radius.
`spectroscopy.fit_agn`	`true`	Include an AGN component in the `XpectraFit` fit.
`spectroscopy.spectrum_fwhm`	2.5	Instrumental resolution (Å) of the spectra.
`spectroscopy.survey_filters`	`[Legacy, SDSS]`	Filter families for synthetic photometry of the spectrum.
`cigale_run` / `cigale_batch`	`false`	Optional additional CIGALE SED fit (any mode); batch variant reuses one model grid for all targets.
`output_dir`	–	Root of the per-target product tree.

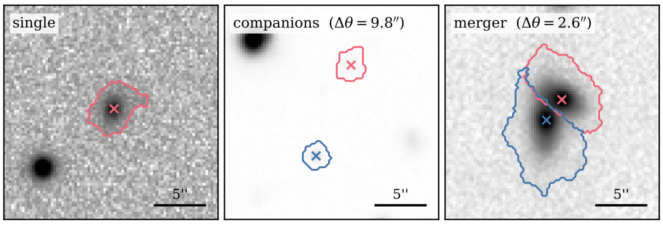

Fig. 2. `xdebpair` source separation on Legacy Survey r-band cutouts for the three classification outcomes of the controlled test set (Sect. 7.1): a single galaxy (left), a resolved pair of companions at $\Delta\theta = 9.8''$ (centre), and an ongoing merger at $\Delta\theta = 2.6''$ (right). Contours show the per-component pixel masks (red: component 1; blue: component 2); crosses mark the detected nuclei. Each panel is 22″ wide.

watershed), identifies the nuclei, and classifies the system as `single`, `companions` (a resolved pair) or `merger` (overlapping isophotes); for two-component systems it partitions the shared footprint between the nuclei with a combination of watershed, two-component PSF fitting with responsibility masks, and random-walker segmentation, iterating to convergence on the mask intersection-over-union. The adapter class `XdebPairAdapter` (in `deblending.py`) exposes the result to the pipeline through a small interface (`masks`, `wcs`, `n_components`, `separation_arcsec`, `classification`) that is deliberately identical to that of the simpler fallback package `xmask`, so the two are interchangeable; a built-in circular stub closes the chain for degraded environments. Figure 2 shows the resulting masks for the three classification outcomes on real Legacy imaging.

4.1.1. Separation policy and blending flag

When the target list carries a morphological `type` column, the `photometry.separation` policy arbitrates between the classification and the pixel-level result: with the default `pair` policy, systems typed `merger`/`pre_merger` keep two components, whereas `post_merger` or other types are forced to a single galaxy (the union of the masks); `central` keeps only the central component and flags the discarded companion; and `total` always measures the mask union as one system.

Independently of the mask geometry, each band b receives a per-band blending flag,

$$\texttt{blend\_flag}(b) = \left[\mathrm{FWHM}_{\mathrm{PSF}}(b) \geq \Delta\theta\right], \tag{1}$$

where $\Delta\theta$ is the nuclear separation measured by `xdebpair`. A flagged band cannot resolve the pair on its own: its per-component fluxes are reported but must be interpreted through the deblending estimators of Sect. 4.4. For the 2.6″ merger of Fig. 2, six of the nine bands are flagged (all but the Legacy grz bands); for the 9.8″ companions, only WISE W4.

4.2. PSF homogenisation and reprojection

All bands are convolved to the coarsest PSF in the set (WISE W4, 12″ FWHM) with Gaussian matching kernels of width $\sigma = \sqrt{\mathrm{FWHM}_{\mathrm{W4}}^2 - \mathrm{FWHM}_b^2} \,/\, (2\sqrt{2\ln 2})$ per band (`convolution.py`); bands already at or above the target resolution are left untouched (no deconvolution is attempted). The actual pixel scale entering the kernel is always read from each image's WCS (`_pixscale_from_wcs`) rather than trusted from the survey registry, so resampled inputs cannot silently corrupt the kernels. Convolved images are then reprojected (`reproject`, bilinear) onto a common TAN grid centred on the target (1.375″ px^{-1}, 45 × 45 px by default).

Two points are worth stressing. First, the full-image convolved and reprojected frames are kept *only* for the visual processing-grid diagnostics; science fluxes are never measured on them. Second, the per-component science measurement uses *isolated* images: each component is cut out of the native frame using its mask, individually convolved to the common PSF and reprojected, so that the PSF wings spread into empty sky instead of into the neighbouring component – close-pair flux is not cross-contaminated by the homogenisation itself (`measure_isolated_photometry` in `photometry.py`).

4.3. Flux measurement and calibration

Within each mask (or aperture), background-subtracted counts are summed and converted to flux density as

$$F_\nu\,[\mathrm{mJy}] = 3631\,\mathrm{Jy} \times 10^{-0.4\,\mathrm{ZP}_b}\, C_b \times 10^3 \times (p/p_0)^2\,, \tag{2}$$

where C_b are the summed counts, ZP_b the AB zero point of Table 1, and the last factor (`zp_grid_correction`) restores the zero-point-native pixel scale p_0 when the image grid has scale $p \neq p_0$: resampling services interpolate pixel values like a surface brightness, so summed counts scale as $(p_0/p)^2$ for a fixed sky aperture (verified empirically by downloading the same fields at several scales). The background level and rms are estimated from sigma-clipped statistics of the sky pixels outside the source masks, using `sep` (Barbary 2016) background maps for well-sampled frames, with coverage gaps (GALEX circular field edges, WISE tile padding) excluded. Flux uncertainties are $\sqrt{N_\mathrm{pix}}\,\sigma_\mathrm{rms}$ converted through the same calibration.

4.4. Deblending of blended bands and forced photometry

For bands flagged by Eq. (1) the pipeline computes three independent estimators of the flux partition between the components (`deblend_photometry.py`): (i) a *ratio prior*, which transports the component flux ratio measured in the unblended optical bands to the blended band; (ii) a *template fit* in the spirit of T-PHOT-like codes, which fits the blended image with the two mask-defined components (and optionally neighbouring sources) convolved to the band's PSF; and (iii) a *combined* estimator merging both. In addition, for the WISE bands the pipeline queries the unWISE forced photometry of the Legacy Surveys DR10 catalogue (Dey et al. 2019; Lang 2014), in which the WISE fluxes are measured by forcing the positions and morphologies of the higher-resolution optical detections – an external, catalogue-level deblending that provides a valuable cross-check of the image-level estimators. All estimators are propagated into the per-target galaxy table so the user can compare them per band.

4.5. Per-component SEDs

The end product of the photometric branch is a long-format table (one row per band) with per-component (`gal1`, `gal2`), total-system, convolved-total and circular-aperture fluxes, blending flags and the deblending estimators, plus per-component SED plots. Figure 3 compares the SEDs of the three systems of Fig. 2: for the resolved pair the two components are measured independently in the unblended bands, while for the 2.6″ merger the optical bands separate the components and the encircled (blended) bands rely on the deblending estimators.

5. Spectroscopic pipeline

In `spectroscopy` (and `both`) mode, the resolved spectrum (Sect. 2.3) is fitted with the companion package `XpectraFit` through the thin adapter `run_spectral_fit` (`spectral_fit.py`). `XpectraFit`'s `XpectraFitter` class orchestrates a full optical spectral analysis:

- *Stellar continuum*: pPXF full-spectrum fitting (Cappellari 2017) with the E-MILES simple stellar population library (Vazdekis et al. 2016), yielding the stellar kinematics ($\sigma_\star$ and its uncertainty), mass-weighted population diagnostics, and fibre-aperture stellar masses;
- *AGN component*: an optional AGN continuum plus broad-line components (enabled by `spectroscopy.fit_agn`), from which the AGN fraction `fAGN` and broad Hα/Hβ-based black hole masses $\log M_\mathrm{BH}$ are derived (e.g. Greene & Ho 2005);
- *Emission lines*: Gaussian measurements of 22 narrow lines (fluxes, equivalent widths, velocity offsets, line widths, signal-to-noise), feeding BPT (Baldwin et al. 1981; Kewley et al. 2001; Kauffmann et al. 2003) and WHAN (Cid Fernandes et al. 2011) classifications;
- *Continuum indices*: D_n4000 (Balogh et al. 1999) and Lick indices;
- *Synthetic photometry*: the observed spectrum (preferred), the best-fit model, or the stellar continuum is integrated through the configured filter families (Legacy and SDSS by default), producing `synth_*` broadband fluxes in the same mJy convention as the imaging pipeline – the key ingredient of the combination step.

The fibre spectrum measures the *total* central light of the system, not one deblended component; the spectral products are therefore always associated with the total aggregate, never with `gal1` or `gal2`.

A reproducibility detail is worth recording. Early batch runs showed run-to-run differences at the last-decimal level in some spectral parameters: the multithreaded BLAS behind pPXF evaluates reductions in a non-deterministic order, and a residual-to-noise ratio lying within floating-point noise of the σ-clipping threshold could be clipped in one run and kept in the next, occasionally toggling entire fit branches. The fix (in `XpectraFit`'s `stellar.py`) rounds the clipping ratio to eight decimals before the threshold comparison, making the clip decision – and thus the full pipeline – bitwise reproducible across runs while leaving the statistics unchanged.

6. Combining photometry and spectroscopy

6.1. Aperture normalisation

In `both` mode the two branches are merged into one flat product per target (`Pipeline._combine_phot_spec`). The physical frame follows Zou et al. (2024, Sects. 3.2–3.3): the fibre spectrum suffers aperture losses, so the imaging photometry – which measures the whole system – is taken as the absolute flux reference and the spectroscopy is corrected to total light. Zou et al. (2024) apply a single grey (r-band) factor; `photextra_pipeline` generalises this to a *wavelength-dependent* correction (`spec_normalization.py`). For every configured band whose effective wavelength lies inside the observed spectral coverage and which has both a total-aperture imaging flux and a synthetic flux, an anchor ratio $F_\mathrm{phot}(b)/F_\mathrm{synth}(b)$ is formed, and a smooth scale

$$\ln s(\lambda) = c_0 + c_1 x + c_2 x^2, \qquad x \equiv \ln(\lambda/\lambda_0), \tag{3}$$

is fitted to the anchors, with λ_0 the geometric mean of the anchor pivot wavelengths. The polynomial degree (0 = grey, 1 = colour slope, 2 = curvature) is selected automatically and conservatively: a degree upgrade requires enough anchors (≥ 3 for degree 1, ≥ 5 for degree 2), sufficient wavelength lever arm, a BIC improvement of at least 6, and a $\geq 5\,\%$ improvement in leave-one-out cross-validation, with 3σ clipping of outlier anchors (at most 34 % clipped) and a condition-number guard on the design matrix. With the three Legacy anchors available today the selected degree is in practice 0 or 1; the scheme automatically gains freedom as more bands (SDSS, HSC, S-PLUS,

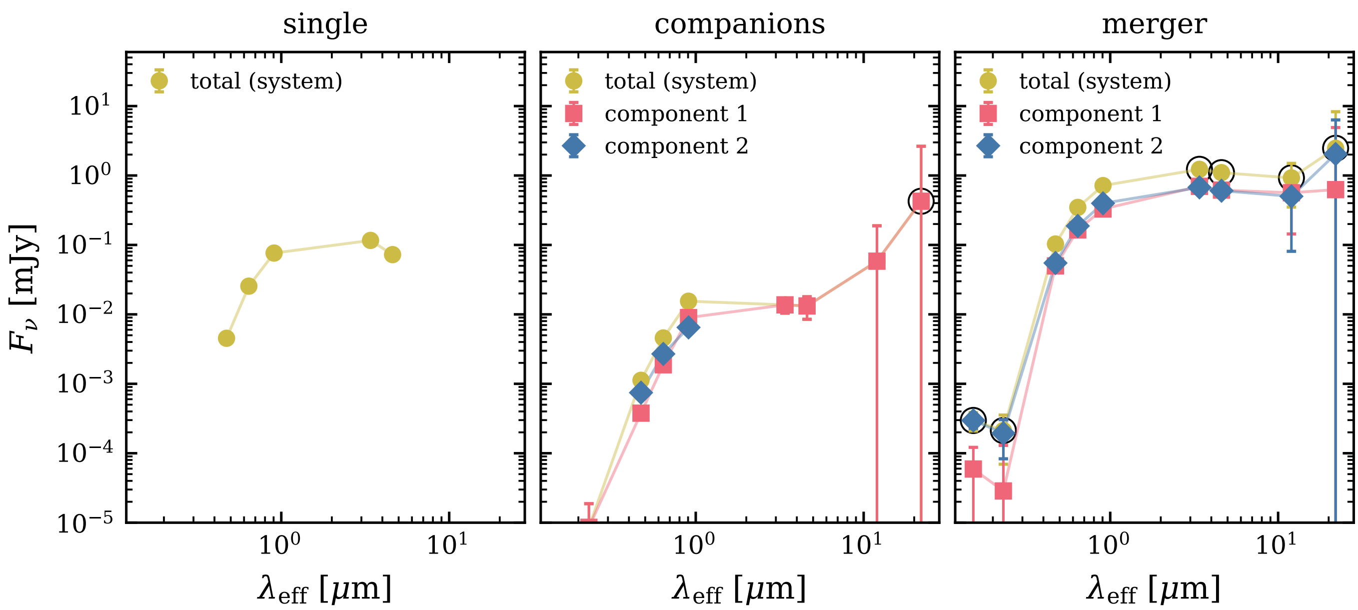

Fig. 3. Per-component SEDs measured by `photextra_pipeline` for the three systems of Fig. 2 (from the pipeline's own photometry products). Yellow circles: total system; red squares / blue diamonds: deblended components. Black open circles mark bands with $\mathrm{FWHM_{PSF}} \geq \Delta\theta$ (Eq. 1), where the per-component split relies on the deblending estimators of Sect. 4.4. Only positive fluxes are shown.

J-PLUS) are added to the configuration – the anchor collection is generic over the survey registry, with no hard-coded band list. The method is adapted from the normalisation introduced in the CIGALE `cigale2s` development branch, reimplemented standalone.

Every synthetic broadband flux is rescaled by $s(\lambda)$ evaluated at its own effective wavelength, and every emission-line flux at its own observed wavelength, capturing differential colour/aperture mismatches a grey factor cannot. For diagnostics, the product also carries the historical estimators: the S/N^2-weighted multi-band grey factor, the single-band (r) factor of Zou et al. (2024), the inter-band dispersion of the grey ratios (values above ∼15–20 % flag strong colour gradients or AGN contamination), and a fibre-matched recalibration factor. The latter uses a second, always-measured set of imaging fluxes through a synthetic aperture matched to the spectrograph fibre (1.5″ DESI, 3″ SDSS diameter); if no total-aperture anchor is usable the pipeline falls back to these fibre fluxes (spectrophotometric recalibration only, flagged `norm_aperture="fiber_fallback"`).

6.2. Rescaled and invariant quantities

Because stellar mass scales linearly with flux at fixed mass-to-light ratio, correcting the fibre-aperture mass after the fit is equivalent to rescaling the spectrum before it (Zou et al. 2024); the product therefore exports

$$M_\star^{\rm total} = M_\star^{\rm fibre} \times s(\lambda_0), \quad (4)$$

(the `stellar_mass_total` column), keeping the original fibre-aperture masses for diagnostics. Deliberately *not* rescaled are: the black hole masses (nonlinear in luminosity, and the unresolved broad-line region is already fully captured by the fibre, so a galaxy-wide aperture ratio would over-correct), the stellar velocity dispersion, D_n4000, `fAGN`, and all flux-ratio classifications (BPT/WHAN), which are aperture-independent. Figure 4 shows the combination product for a representative cluster galaxy.

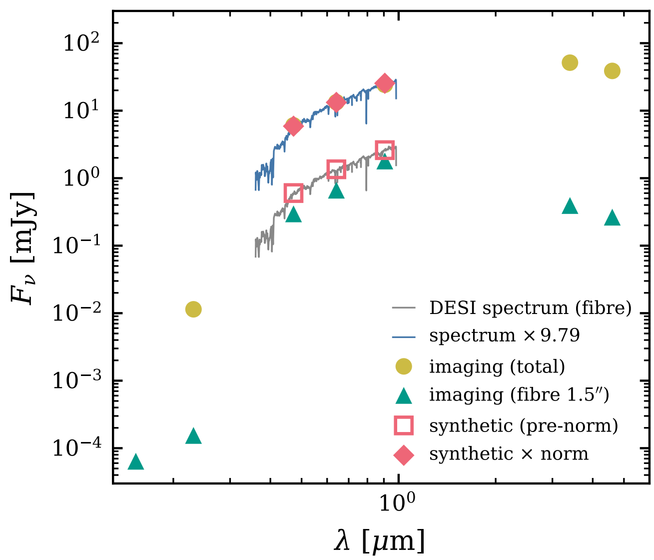

Fig. 4. Photometry–spectroscopy combination (`mode="both"`) for a representative MKW 8 member (DESI targetid 2842593834565632, $z = 0.0279$). The observed DESI fibre spectrum (grey) under-captures the total light measured by the imaging photometry (yellow circles) but matches the fibre-aperture imaging fluxes (green triangles). The fitted normalisation $s(\lambda)$ (here ≃ 9.8 at λ_0, anchored on the Legacy g, r, z bands) rescales the spectrum (blue) and its synthetic photometry (open → filled red symbols) to the total-light scale.

Table 3. Controlled five-system test set (test_output_xdeb).

System	xdebpair class	$\Delta\theta$ ($''$)	Blended bands$^{(a)}$
merger_A	merger	2.6	6/9
merger_B	companions	9.8	1/9
merger_C	merger	4.5	6/9
merger_D	single	–	0/9
merger_E	single	–	0/9

Notes. $^{(a)}$ Bands with $\mathrm{FWHM_{PSF}} \geq \Delta\theta$ (Eq. 1).

6.3. Optional CIGALE SED fitting

Independently of the built-in combination, the pipeline can drive CIGALE (Boquien et al. 2019) as an additional SED fit in any mode (cigale_integration.py): in both mode with the fixed spectroscopic redshift, broadband fluxes and the full DESI spectrum (CIGALE's native spectrophotometric mode); in spectroscopy mode on the spectrum alone; and in photometry mode with the redshift left free, so CIGALE χ^2-scans its redshift grid and returns a photometric redshift (cigale_z_phot). A batch variant builds a single multi-target input so the (expensive) model grid is computed only once. The resulting cigale_* columns (stellar mass, SFR, χ^2) are appended to the per-target products.

7. Validation

7.1. Controlled close-pair test set

The deblending-photometry chain was validated on five real systems chosen to span the classification space (Table 3); three of them are shown in Figs. 2 and 3. The classifications and separations are stable under re-runs, and the per-band blending flags follow Eq. (1): the 2.6$''$ and 4.5$''$ mergers are resolved only by the Legacy optical bands (six of nine bands flagged), the 9.8$''$ companion pair is blended only in WISE W4, and the single galaxies are never flagged. In the resolved pair the independent component SEDs recover two galaxies of comparable optical flux with diverging mid-infrared behaviour, while in the close merger the optical-band component split propagates consistently into the deblending estimators of the flagged bands.

7.2. Production run: 450 members of MKW 8

As an end-to-end test at survey scale, the pipeline was run in both mode on 450 spectroscopic members of the poor cluster MKW 8 ($z \simeq 0.027$; $0.023 \leq z \leq 0.033$) from the CHANCES target base, using seven imaging bands (GALEX FUV/NUV, Legacy grz, WISE W1/W2) and automatic DESI spectrum resolution. The outcome:

- 446/450 targets (99.1 %) produced complete combined products; all 446 spectra were resolved from DESI (SPARCL) and all 446 spectral fits converged (spec_status = ok);
- the 4 failures were all external and transient (unWISE forced photometry queries exhausting their retries against a rate-limited VizieR mirror); they are recorded in failed_targets.jsonl and recoverable by re-running, at which point the checkpointing reuses everything already computed;
- the median end-to-end cost was ≃221 s per target (interquartile range 181–255 s, dominated by the pPXF+AGN spectral fit and the image downloads), while re-entering the run with cached photometric and spectral products costs a median of 0.6 s per target for the combination step alone;
- the total-aperture normalisation succeeded for all 446 galaxies (norm_aperture = total, three Legacy anchors each), with a median aperture correction $s(\lambda_0) = 6.4$ (16th–84th percentiles: 3.3–14.1) – consistent with a 1.5$''$ DESI fibre sampling the core of $z \simeq 0.027$ galaxies – and a median inter-band ratio dispersion of 11 %.

Figure 5 summarises the distribution of aperture corrections and the WHAN classification mix of the sample (126 star-forming, 72 sAGN, 31 wAGN, 100 retired, 35 passive, 82 unclassifiable), with the BPT diagram yielding a consistent picture (109 SF, 73 composite, 53 LINER, 47 Seyfert); a red-sequence-dominated population with a substantial retired/passive fraction, as expected for cluster members with a median D_n4000 of 1.65.

7.3. Quality assurance: the WISE zero-point case

We report one calibration bug found and fixed during development because it illustrates the value of physically motivated validation at the pipeline level. The flux conversion of Eq. (2) assumes AB zero points uniformly, but the WISE Atlas zero points originally entered in the survey registry were the native *Vega* magnitude zero points. The resulting W1/W2 fluxes were inflated by roughly an order of magnitude – the $10^{0.4\Delta m}$ of the Vega→AB offsets, a factor ≃12 in W1 and ≃22 in W2. The error was caught not by unit tests but by a physical sanity check: for normal stellar photospheres, F_ν must decline from the 1.6 μm stellar bump into the mid-infrared (the Rayleigh–Jeans tail), whereas the measured WISE-to-optical flux ratios of ordinary early-type cluster members sat an order of magnitude above that expectation. The fix converts the WISE zero points to AB equivalents once, in the registry, using the offsets of Jarrett et al. (2011) (Table 1), so the single AB conversion of Eq. (2) applies uniformly to all bands. The episode motivated the permanent inclusion of SED-level ratio diagnostics in the validation plots, and – together with the deterministic σ-clipping fix of Sect. 5 and the refuse-to-degrade download policy of Sect. 3 – reflects the pipeline's general design stance: failures and calibration errors must be loud, recorded and physically checkable, never silent.

8. Code availability

photextra_pipeline is written in Python (≥3.8) and depends only on standard open-source infrastructure: numpy, scipy, astropy (Astropy Collaboration et al. 2022), matplotlib, photutils, sep (Barbary 2016), reproject, astroquery (Ginsburg et al. 2019) and pyyaml; the spectral branch additionally requires ppxf (Cappellari 2017) and the SPARCL client (Juneau et al. 2024), and the optional SED fitting requires CIGALE (Boquien et al. 2019). The package installs with pip and exposes both a Python API (Pipeline) and a command-line entry point (photextra) driven entirely by the YAML configuration of Table 2. The source code of photextra_pipeline and of the companion packages xdebpair, xmask and XpectraFit will be released on GitHub with an archived version on Zenodo upon publication [repository URL, DOI and license to be finalised; license placeholder: MIT]. The per-target products used in Sect. 7 are available from the author on request.

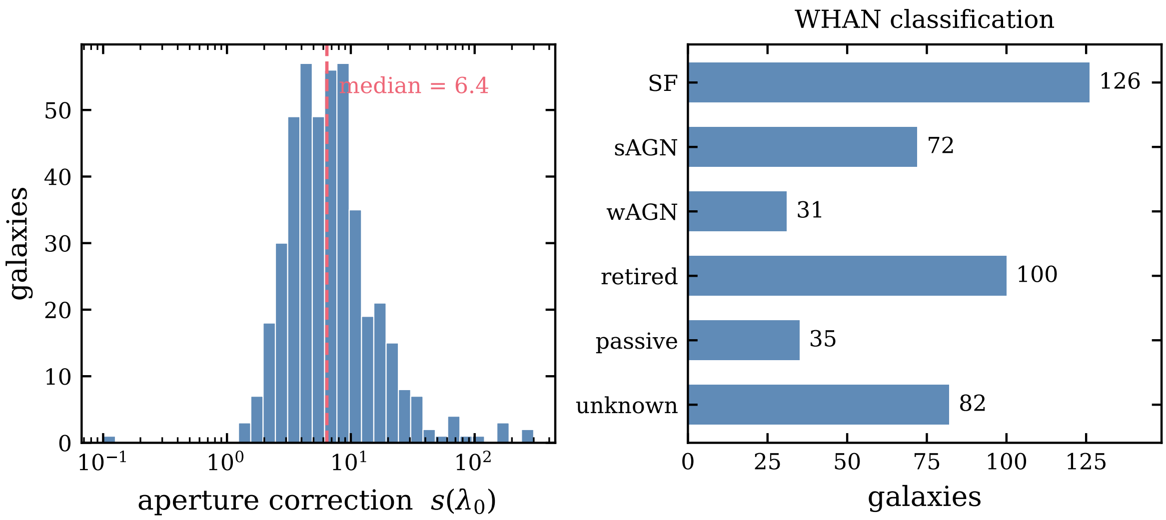

Fig. 5. MKW 8 production run (446 combined products). Left: distribution of the applied aperture correction $s(\lambda_0)$ (fibre to total light; median 6.4, dashed line). Right: WHAN emission-line classification of the sample as measured by `XpectraFit`.

9. Summary

`photextra_pipeline` turns a target list of (ID, RA, Dec, z) into component-level, PSF-homogenised, nine-band SEDs and aperture-corrected spectroscopic parameters through a single configuration-driven, checkpointed pipeline. Its distinguishing features are (i) pixel-level deblending of close and merging pairs (`xdebpair`) propagated consistently through PSF homogenisation via isolated-component photometry and through per-band blending flags; (ii) automatic spectrum resolution (DESI/SDSS) and a full stellar+AGN spectral analysis (`XpectraFit`); (iii) a wavelength-dependent, conservatively regularised normalisation of the fibre spectroscopy to total light, extending the grey aperture correction of Zou et al. (2024); and (iv) survey-scale robustness demonstrated on 450 cluster members with a 99.1 % success rate and second-scale resume cost. The pipeline is being applied to the CHANCES cluster survey and is designed to absorb additional imaging surveys by configuration alone.

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References

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167
Baldwin, J. A., Phillips, M. M., & Terlevich, R. 1981, PASP, 93, 5
Balogh, M. L., Morris, S. L., Yee, H. K. C., Carlberg, R. G., & Ellingson, E. 1999, ApJ, 527, 54
Barbary, K. 2016, Journal of Open Source Software, 1, 58
Bertin, E. & Arnouts, S. 1996, A&AS, 117, 393
Boquien, M., Burgarella, D., Roehlly, Y., et al. 2019, A&A, 622, A103
Cappellari, M. 2017, MNRAS, 466, 798
Chambers, K. C., Magnier, E. A., Metcalfe, N., et al. 2016, arXiv:1612.05560
Cid Fernandes, R., Stasińska, G., Mateus, A., & Vale Asari, N. 2011, MNRAS, 413, 1687
DESI Collaboration, Adame, A. G., Aguilar, J., et al. 2024, AJ, 168, 58
Dey, A., Schlegel, D. J., Lang, D., et al. 2019, AJ, 157, 168
Ginsburg, A., Sipőcz, B. M., Brasseur, C. E., et al. 2019, AJ, 157, 98
Greene, J. E. & Ho, L. C. 2005, ApJ, 630, 122
Jarrett, T. H., Cohen, M., Masci, F., et al. 2011, ApJ, 735, 112
Juneau, S., Olsen, K. P., Nikutta, R., et al. 2024, AJ, 167, 258
Kauffmann, G., Heckman, T. M., Tremonti, C., et al. 2003, MNRAS, 346, 1055
Kewley, L. J., Dopita, M. A., Sutherland, R. S., Heisler, C. A., & Trevena, J. 2001, ApJ, 556, 121
Lang, D. 2014, AJ, 147, 108
Martin, D. C., Fanson, J., Schiminovich, D., et al. 2005, ApJ, 619, L1
McGlynn, T., Scollick, K., & White, N. 1998, in IAU Symp. 179, New Horizons from Multi-Wavelength Sky Surveys, ed. B. J. McLean et al. (Dordrecht: Kluwer), 465
Skrutskie, M. F., Cutri, R. M., Stiening, R., et al. 2006, AJ, 131, 1163
Sifón, C., et al. 2025, A&A, in press (CHANCES survey description)
Vazdekis, A., Koleva, M., Ricciardelli, E., Röck, B., & Falcón-Barroso, J. 2016, MNRAS, 463, 3409
Wright, E. L., Eisenhardt, P. R. M., Mainzer, A. K., et al. 2010, AJ, 140, 1868
York, D. G., Adelman, J., Anderson, J. E., et al. 2000, AJ, 120, 1579
Zou, F., Brandt, W. N., Ni, Q., et al. 2024, ApJ, 961, 173